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The Messy Middle: Negotiating Real-World School-Industry Partnerships



Slides and Resources

https://tinyurl.com/NDCTE26

Tony Perry

08/03/2026

NDCTE

By the end of this session…

- Know how to **apply systems thinking** to WBL and industry partnerships
- Evaluate evidence of **strategies to create and sustain local partnerships** with industry
- **Develop an action plan** to create or strengthen partnerships in your community

The Big Ideas

Lasting partnerships are built on **trust** between students, school officials, and employers

Trust builds in the system over time; **student-employer** interactions build trust

How you do this requires local expertise

Join the Vevox session

Go to **vevox.app**

Enter the session ID:
453-037-004

Or scan the QR code



About me

Create LAB

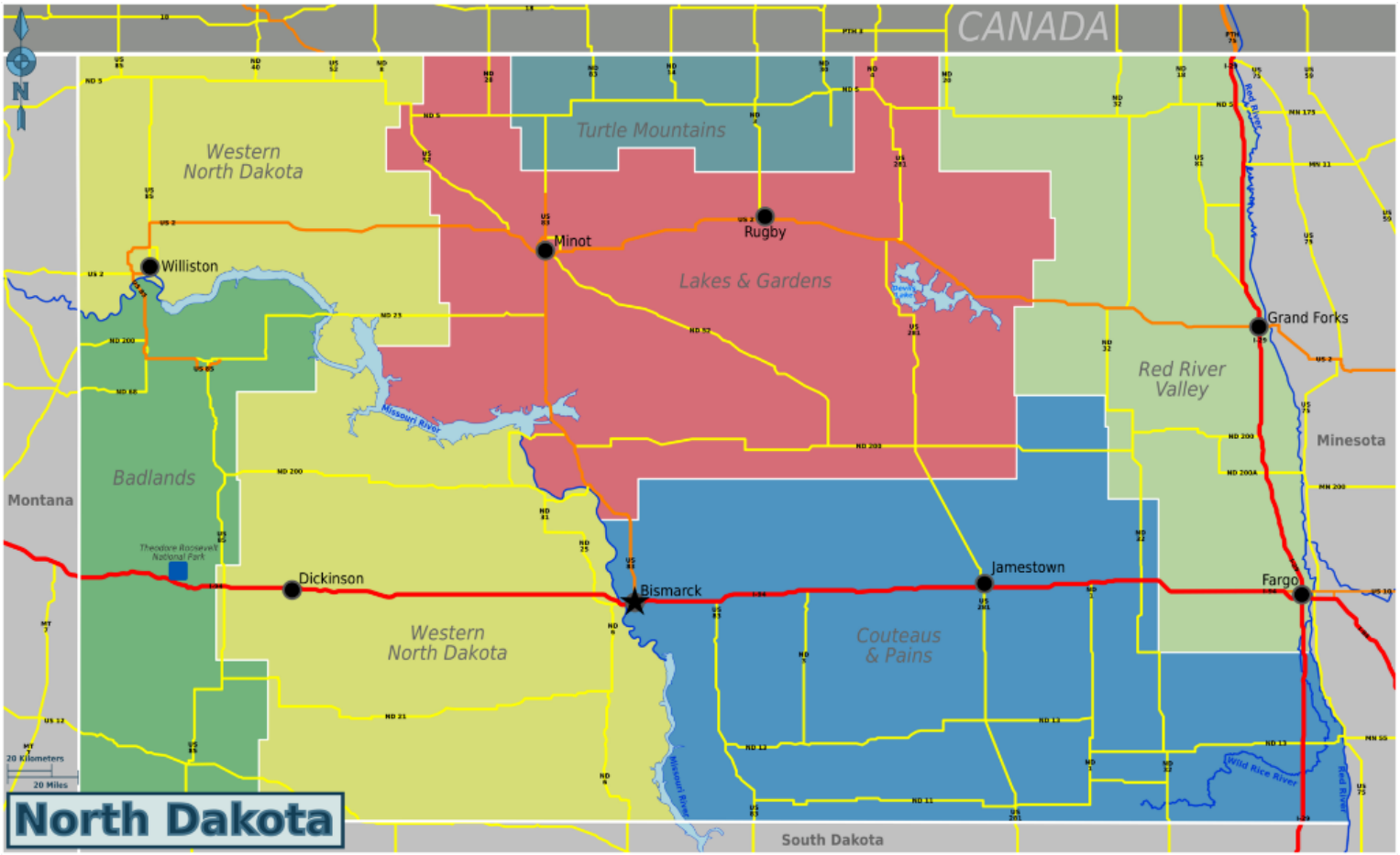
Assistant Professor, STEM Education, University of North Dakota

15+ years experience developing and supporting innovative technology education learning experience.

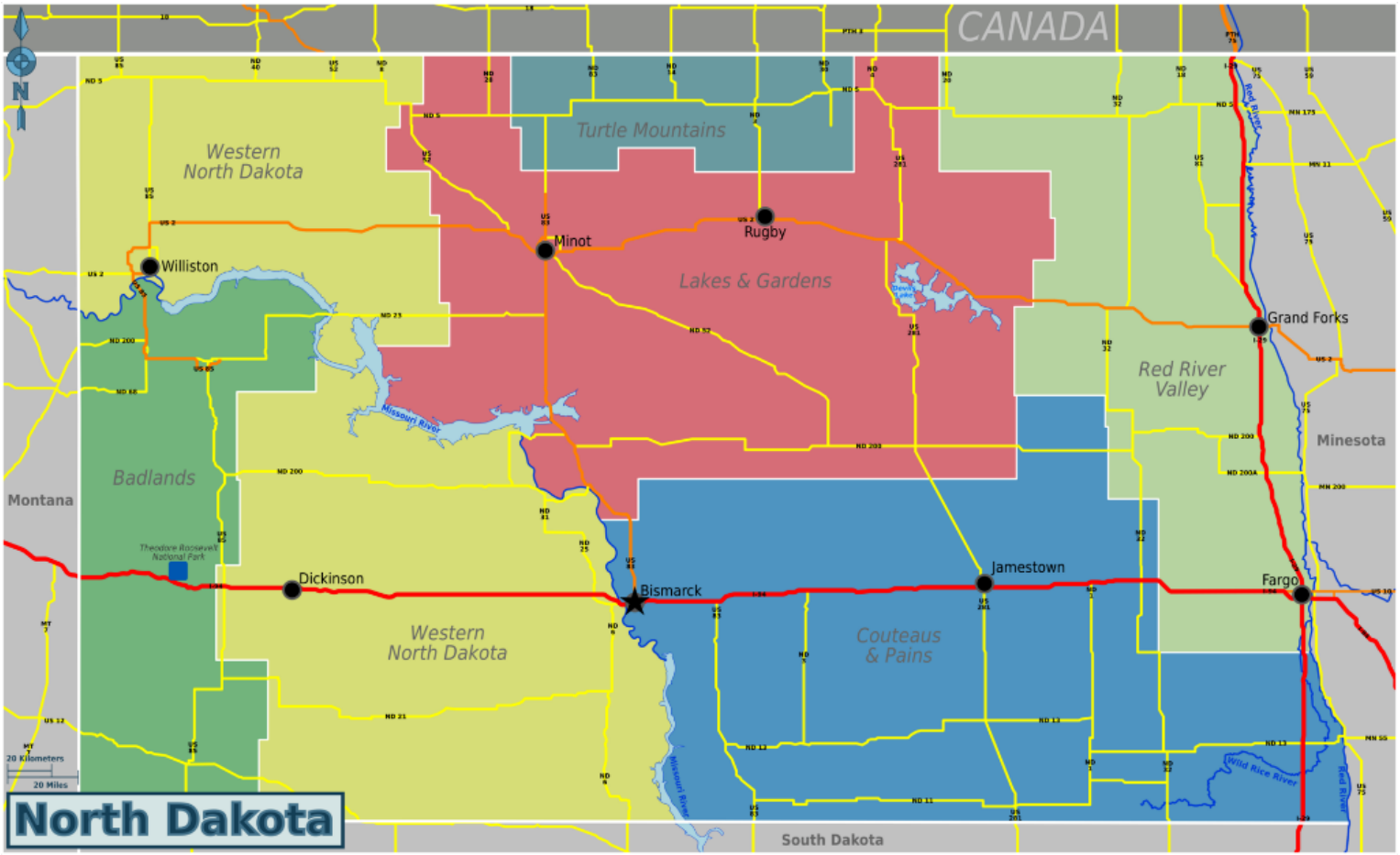
Research Interests: CTE, improving systems and student pathways into emerging technology

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Pin your location on the map



Pin your location on the map



Join at: **vevox.app**

ID: **453-037-004**

Question slide

My primary role is

Classroom Teacher

0%

Career Counselor

0%

Administrator

0%

Other

0%

My primary role is



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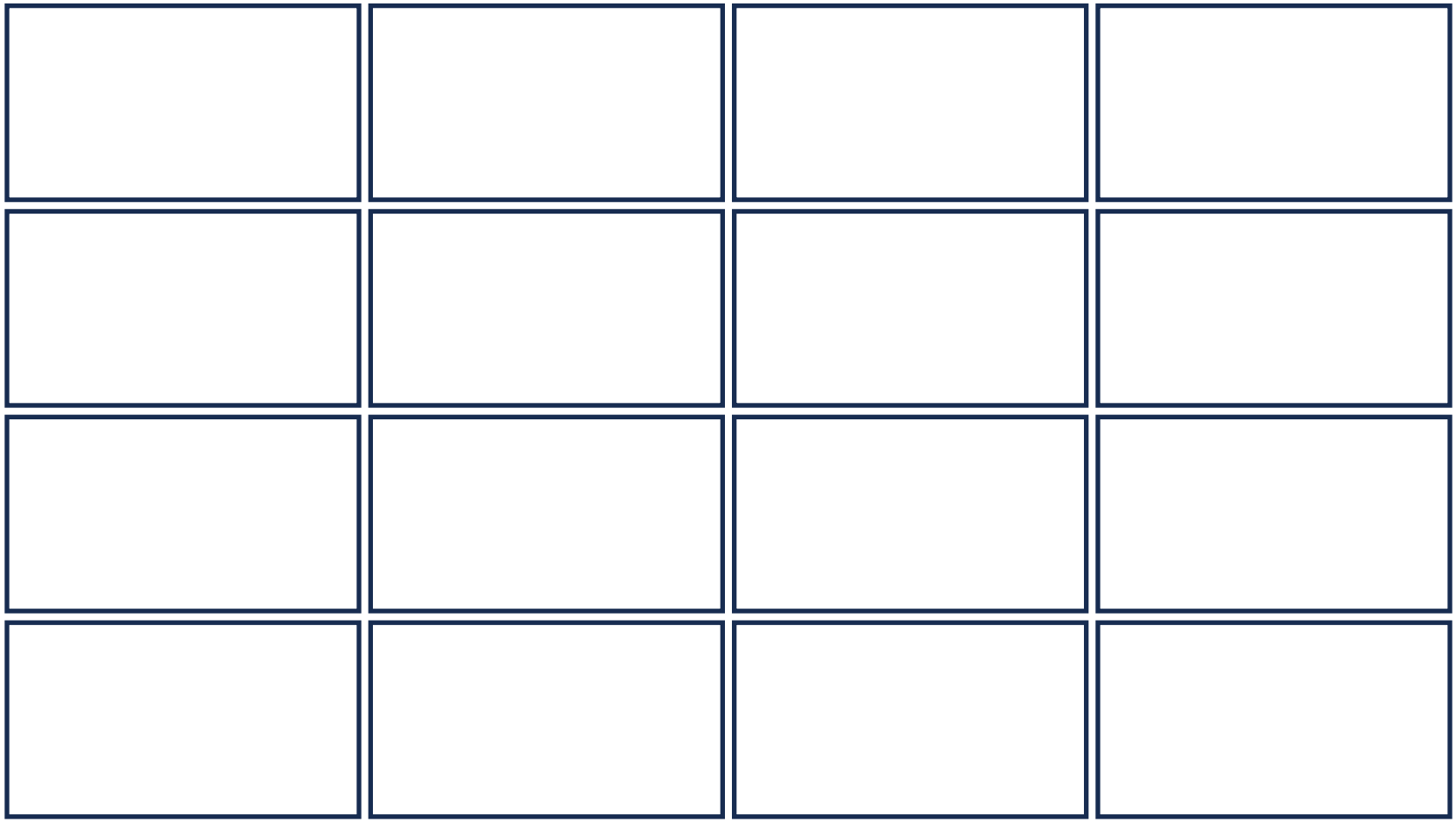
My biggest challenge when working with industry partners is...

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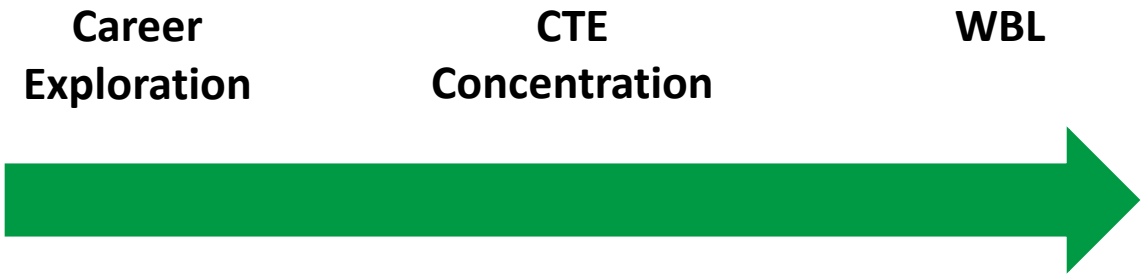
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Results slide

My biggest challenge when working with industry partners is...



Mutually Beneficial Partnerships on the WBL Continuum



Where do we begin to build school-industry partnerships?

WBL Placements?
Advisory Boards?
CLNA?
Something else?

Barriers to partnership

- Most partnerships are **one-sided** (Strobel & Sun, 2020)
- Barriers to partnership (Rios et al., 2014)
  - Structural
  - Cultural
  - Procedural
  - Motivational
- Educators are trained as instructional leaders, not cross-sector organizers (Epstein, 2018)

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Partnerships over Placements

- Center student experiences, not institutional goals
- Leverage complementary assessments of school-based and industry personnel

Mutually Beneficial School-Industry Partnerships (Perry, 2026) **allow…**

- Scale and sustainability
- Emergence of new and innovative practices across the WBL continuum
- Equitable access

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Research Data

“Reverse Engineer” the systems of three Existence Proofs

- School-wide experiences
- Evidence of positive student impact
- At least 5 years running
- urban, suburban, rural

What design features do these cases have in common to **grow partnerships over time?**

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Existence Proofs

	Mid-America STEM	Northeast Rural	Innovative STEM
Student Experience	PrBL	2 days/week offsite WBL	"Flipped" Internship
Locale	Urban	Rural	Suburban
Region	Midwest	Northeast	New England
Type	Career Academy	"alternative" school	School-within-a-school
Size	~300	~60	~120
Population	2/3 male, otherwise matches district	~1/2 from Tribal Lands	More URM than student population

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Common Design Elements

- Explicit trust-building

- Signals of institutional credibility

- (Striving for) robust systems

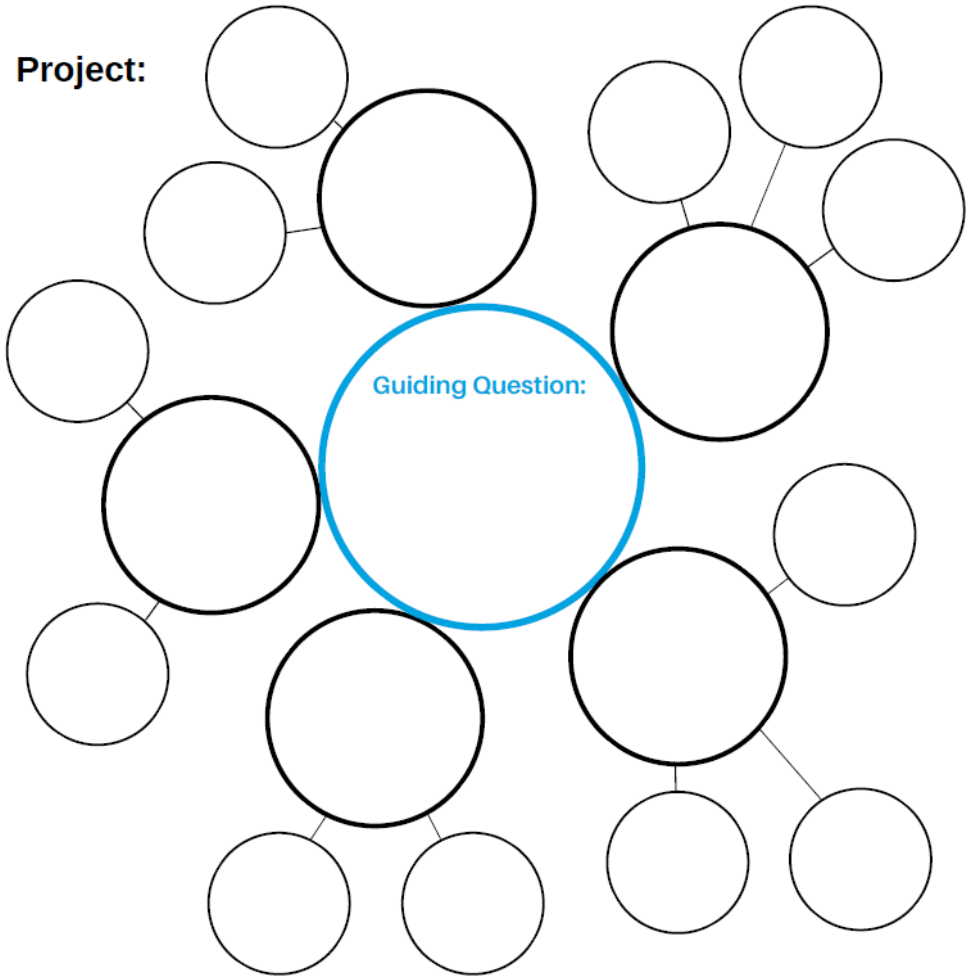
- Aligning instructional subsystems with industry practices

Asset Mapping

Identify an existing program or project where you want to grow existing partners

Big Circles: Themes

Small Circles: specific companies/people



https://workforce.education.asu.edu/wp-content/uploads/2024/07/Community-educator-asset-map-v1.pdf

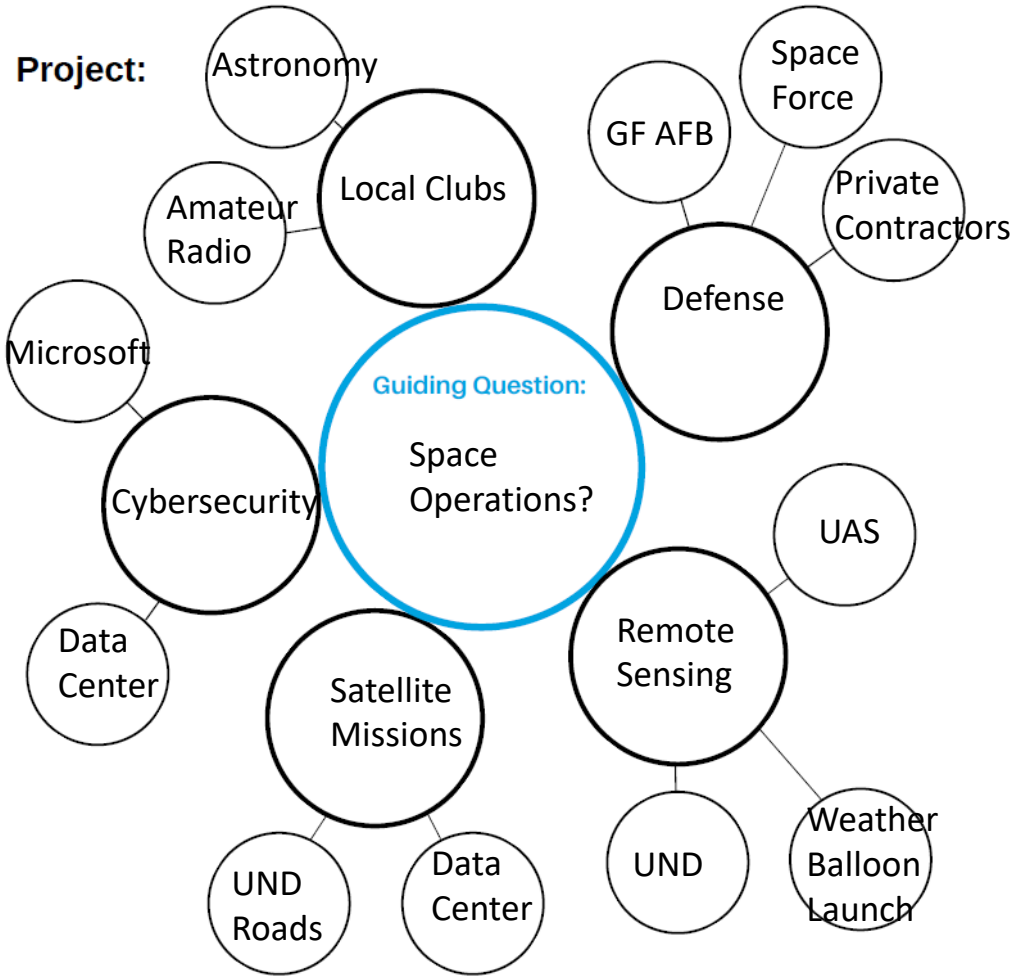
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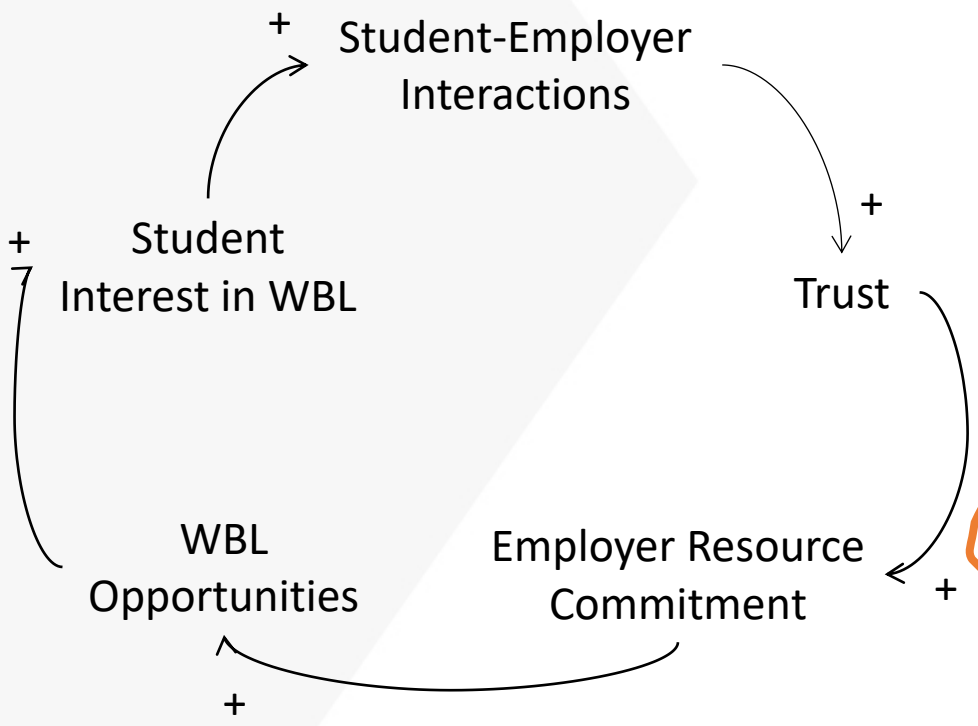
To-Do

List local employers that you would like to be more engaged with your CTE program. These could be new employers or existing partners you would like additional engagement with.

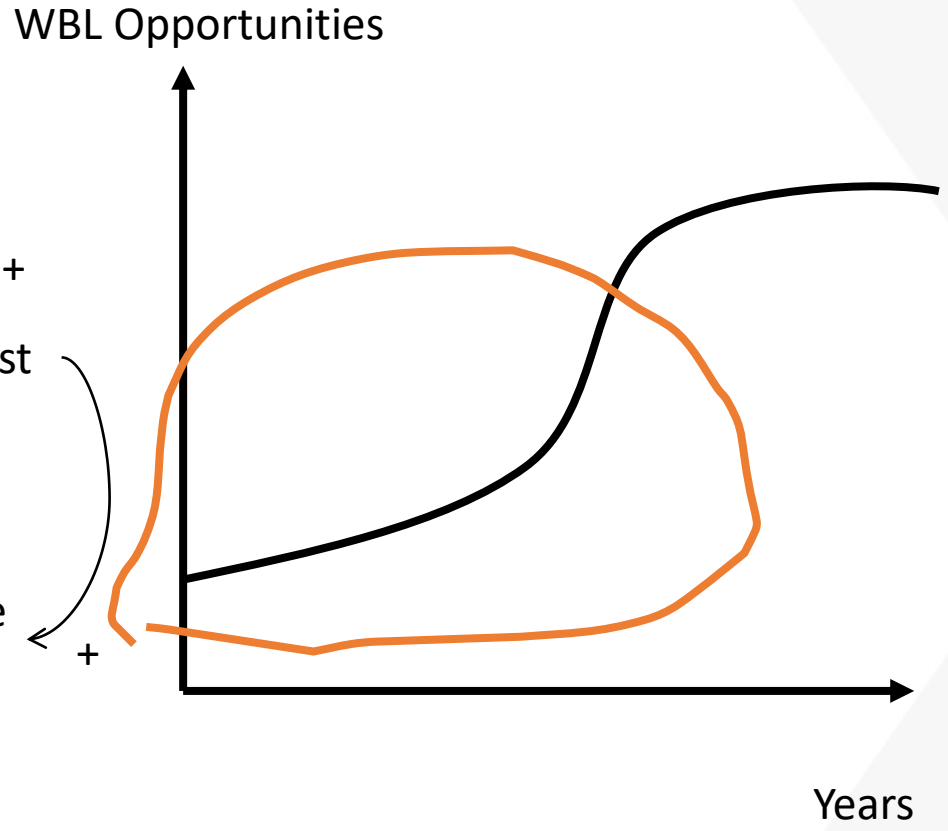
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Thinking in Systems

Causal Loop Diagram



Behavior over Time



Explicit trust-building

"[students] light up when another person totally outside of the education system, **or at least outside of the education system they know**, has an adult conversation with them."

"I think [STEM professionals] get inspired by my students. They always come away being excited with how mature and professional and creative the kids are. **they're not like what they expected**"

-Mid-America STEM Classroom Teacher

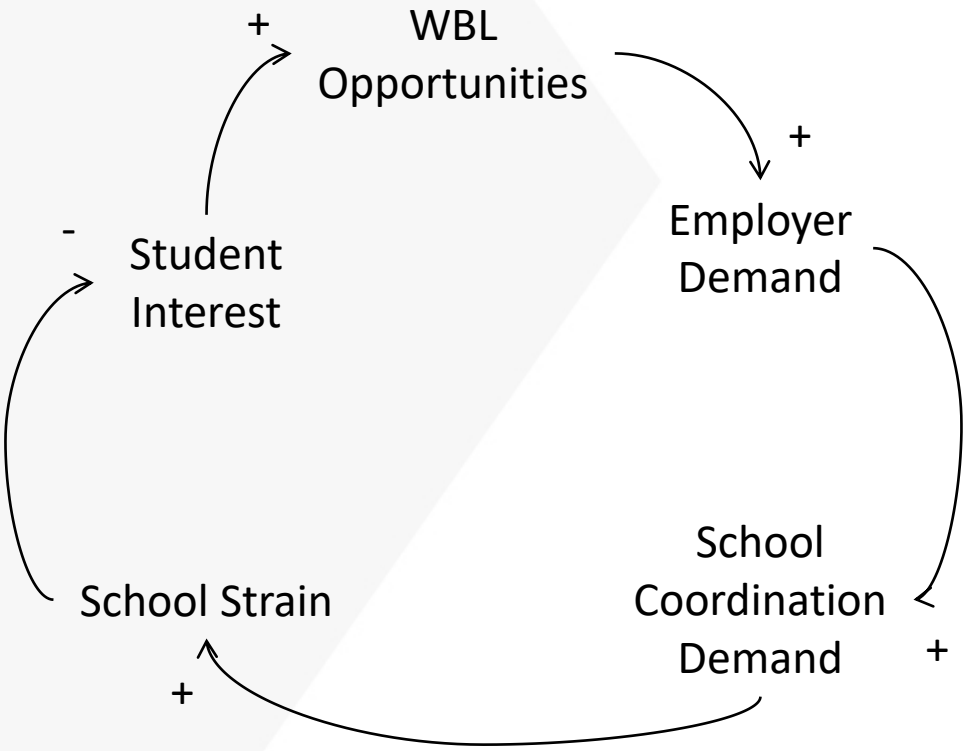
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To-Do

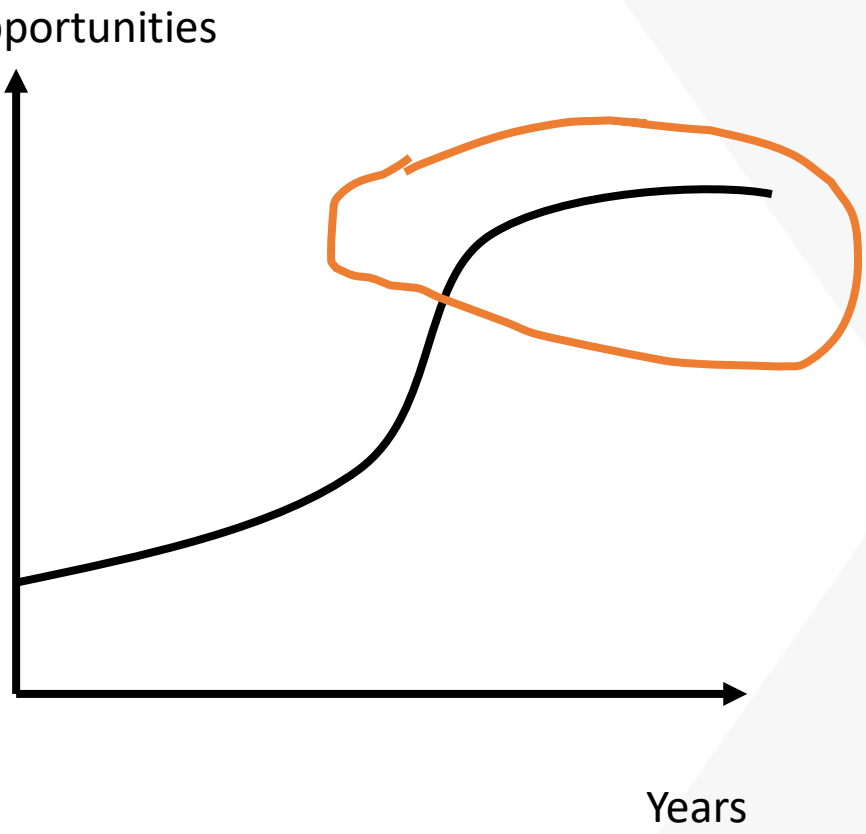
In your current role, how can you facilitate student-industry connections?

Thinking in Systems

Causal Loop Diagram



Behavior over Time



Aligning Instructional Subsystems with Industry

Mid-America	Northeast	Innovative STEM
Flexible Friday for collaboration	Fund student transportation	"Flipped Model" with lunch-time synchronous
Engineer on loan for one school year	Backward map WBL -> learning outcomes	Curriculum co-development w/industry partner
Hosting teacher PD at industry site	Create student opportunities on-site to fill in gaps	

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Example: NIHF STEM School, Akron, OH

Career Exploration	CTE Concentration	WBL
Partner-driven PrBL	Three career academies	Student-driven PrBL w/industry mentors
Common design framework across curriculum, start in MS	Student project showcase nights	
“Flex Friday”	Engineer on “loan”	
PL for educators w/industry partner		

To-Do

Who help manage industry partnerships in the school? How can you reduce the strain on schools?

Next Steps to Growth

Guiding Questions

1) How can I put students, potential partners, and teachers in the same space?
2) Describe the student experience?
3) Industry partner assets?
4) Teacher assets?
5) What is our plan to reconnect?

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Discussion

"…I believe in that effort, that interface, the one-on-one contact, that's the difference-maker relationship."

Manufacturing Plant Manager w/Mid-America STEM

Explicit trust-building

student-school-industry connections

Signals of institutional credibility

Right time for the CEO/Superintendent

(Striving for) robust systems

Focus on complementary assets

Aligning instructional subsystems with industry practices

Start with little changes

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Acknowledgements

Dick Larson, MIT IDSS

"Never let reality stand in the way of what is possible."

https://tinyurl.com/NDCTE26

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anthony.m.perry@UND.edu

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